SANJIVANI COLLEGE OF ENGINEERING,

KOPARGAON

(An autonomous Institute affiliated to SPPU Pune)

Department of Information Technology

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Foundation of Data Science (PEIT-311B)

CIA Part B Activity Report

Topic-

Exploratory Data Analysis of AI_Job_Market & Salary Trends_2025

Group-07

Name	PRN
Ms. Shinde Rachana Nitin	0124UITF1059
Ms. Shete Anushka Pravin	0124UITF1061
Mr. Chavan Varad Subhash	0124UITM1062
Ms. Shinde Sahyadri Bhausaheb	0124UITF1063

Course In-charge :- **Mr. U. B. Sangule**

INTRODUCTION

Exploratory Data Analysis (EDA) is the process of analyzing a dataset to understand its structure, identify missing values, summarize numerical information, and discover patterns through graphs. This project analyzes the AI job market and salary trends for 2025 using Python libraries such as Pandas, Matplotlib, and Seaborn. The dataset is used to explore AI-related job roles, salary distributions, in-demand skills, and job market trends across different countries. The project includes manual EDA and automated EDA using YData Profiling.

Dataset Description

Dataset: AI Job Market & Salary Trends 2025

Dataset Selected CSV file: AI Job Market & Salary Trends 2025.csv

The dataset contains information related to AI job roles, salaries, experience levels, and employment trends. The following columns can be described based on the actual dataset structure:

Column	Description
Job Title	Name of the job role
Salary	Salary offered for the job
Experience Level	Required professional experience
Employment Type contract	Type of employment, such as full-time or
Company Location	Country where the company is located
Employee Residence	Country where the employee resides
Company Size	Size of the company
Required Skills	Skills needed for the job role

Manual Exploratory Data Analysis Python Code

```
# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 2. Load Dataset
df = pd.read_csv("AI Job Market & Salary Trends 2025(1).csv")
df.columns = df.columns.str.strip()
print("First 5 Records:\n", df.head())
print("\nDataset Shape:", df.shape)
print("\nColumn Names:", df.columns.tolist())
df.info()

# 3. Statistical Summary
print("\nStatistical Summary:\n", df.describe(include="all"))

# 4. Identify Columns
num = df.select_dtypes(include=np.number).columns
cat = df.select_dtypes(include="object").columns
print("\nNumerical Columns:", list(num))
print("Categorical Columns:", list(cat))

# 5. Missing Values
print("\nMissing Values:\n", df.isnull().sum())
print("\nMissing Percentage:\n", df.isnull().mean() * 100)

# 6. Handle Missing Values
for c in num:
```

```
    df[c] = df[c].fillna(df[c].median())

for c in cat:
    if df[c].isnull().sum() > 0 and not df[c].mode().empty:
        df[c] = df[c].fillna(df[c].mode()[0])

# 7. Remove Duplicates
print("\nDuplicate Rows:", df.duplicated().sum())
df.drop_duplicates(inplace=True)
print("Shape After Cleaning:", df.shape)

# 8. Unique Values and Categories
for c in cat:
    print("\nValue Counts:", c)
    print(df[c].value_counts().head(10))

# 9. Bar Charts
for c in cat:
    df[c].value_counts().head(10).plot(kind="bar", figsize=(8, 5))
    plt.title("Distribution of " + c)
    plt.xlabel(c)
    plt.ylabel("Count")
    plt.xticks(rotation=45, ha="right")
    plt.tight_layout()
    plt.show()

# 10. Histograms
for c in num:
    plt.figure(figsize=(8, 5))
    plt.hist(df[c].dropna(), bins=20, edgecolor="black")
    plt.title("Distribution of " + c)
    plt.xlabel(c)
    plt.ylabel("Frequency")
    plt.tight_layout()
```

```
    plt.show()

# 11. Box Plots and Outlier Detection
for c in num:
    plt.figure(figsize=(8, 4))
    plt.boxplot(df[c].dropna(), vert=False)
    plt.title("Boxplot of " + c)
    plt.xlabel(c)
    plt.show()

    q1, q3 = df[c].quantile([0.25, 0.75])
    iqr = q3 - q1
    outliers = df[(df[c] < q1 - 1.5 * iqr) |
                  (df[c] > q3 + 1.5 * iqr)]
    print(c, "- Outliers:", len(outliers))

# 12. Correlation Analysis and Heatmap
if len(num) >= 2:
    corr = df[num].corr()
    print("\nCorrelation Matrix:\n", corr)
    plt.figure(figsize=(10, 7))
    plt.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1)
    plt.colorbar(label="Correlation")
    plt.xticks(range(len(num)), num, rotation=90)
    plt.yticks(range(len(num)), num)
    for i in range(len(num)):
        for j in range(len(num)):
            plt.text(j, i, f"{corr.iloc[i, j]:.2f}",
                     ha="center", va="center")
    plt.title("AI Job Market Correlation Heatmap")
    plt.tight_layout()
    plt.show()
print("\nMANUAL EDA COMPLETED SUCCESSFULLY")ESSFULLY")
```

Manual EDA Output

DATASET LOADED SUCCESSFULLY

First 5 Records:

	job_id	job_title	salary_usd	job_des_length	benefits_score	company_name
0	AI00001	AI Research Scientist	90376 ...	1076	5.9	Smart Analytics
1	AI00002	AI Software Engineer	61895 ...	1268	5.2	TechCorp Inc
2	AI00003	AI Specialist	152626 ...	1974	9.4	Autonomous Tech
3	AI00004	NLP Engineer	80215 ...	1345	8.6	Future Systems
4	AI00005	AI Consultant	54624 ...	1989	6.6	Advanced Robotics

[5 rows x 19 columns]

Dataset Shape:

(15000, 19)

Column Names:

['job_id', 'job_title', 'salary_usd', 'salary_currency', 'experience_level', 'employment_type', 'company_location', 'company_size', 'employee_residence', 'remote_ratio', 'required_skills', 'education_required', 'years_experience', 'industry', 'posting_date', 'application_deadline', 'job_description_length', 'benefits_score', 'company_name']

Statistical Summary:

	job_id	job_title	salary_usd	job_des_length	benefits_score	company_name
count	15000	15000	15000.000000	15000.000000	15000.000000	15000
unique	15000	20	NaN	NaN	NaN	16
top	NaN	NaN	NaN	NaN	NaN	TechCorp Inc
freq	1	808	NaN	NaN	NaN	980
mean	NaN	NaN	115348.965133	1503.314733	7.504273	NaN
std	NaN	NaN	60260.940438	576.127083	1.450870	NaN
min	NaN	NaN	32519.000000	500.000000	5.000000	NaN
25%	NaN	NaN	70179.750000	1003.750000	6.200000	NaN

50%	NaN	NaN	99705.000000	1512.000000	7.500000	NaN
75%	NaN	NaN	146408.500000	2000.000000	8.800000	NaN
max	NaN	NaN	399095.000000	2499.000000	10.000000	NaN

[11 rows x 19 columns]

Missing Values:

job_id 0

job_title 0

salary_usd 0

salary_currency 0

experience_level 0

employment_type 0

company_location 0

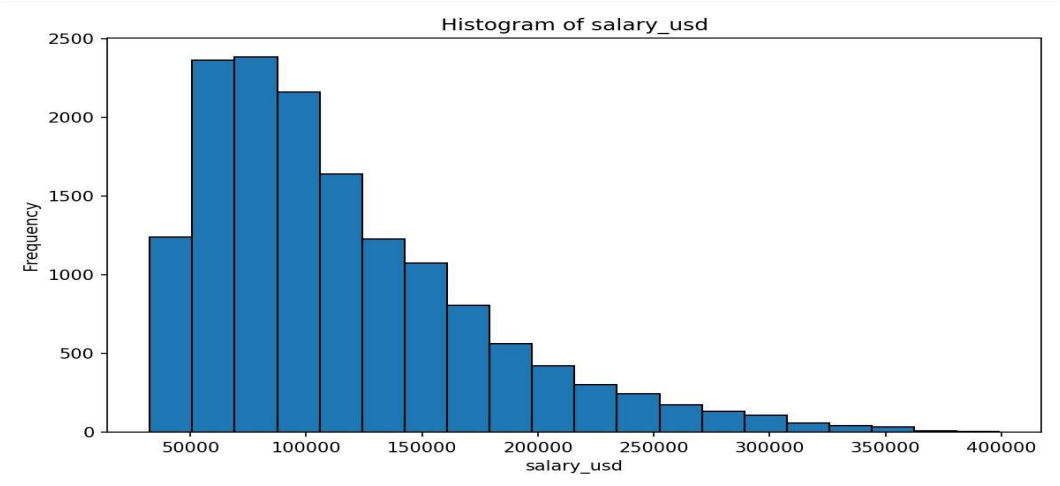
company_size 0

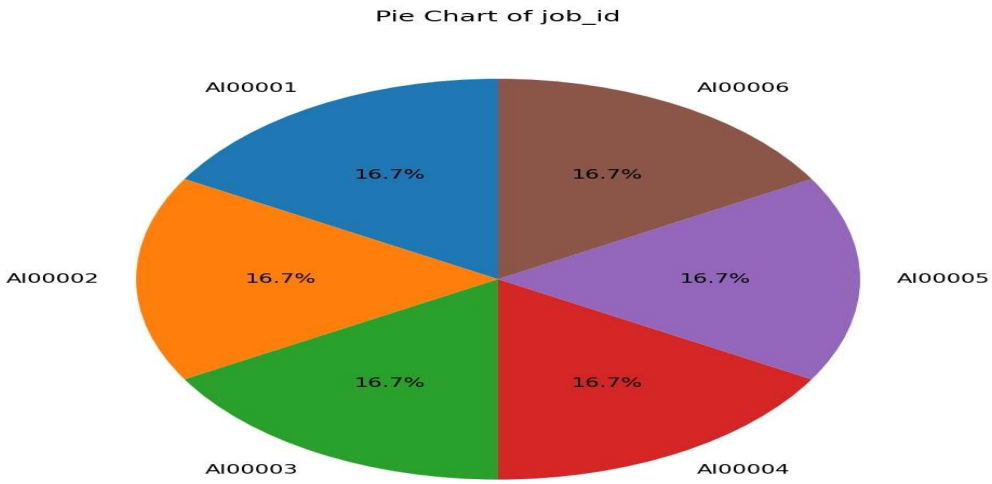
Duplicate Rows: 0

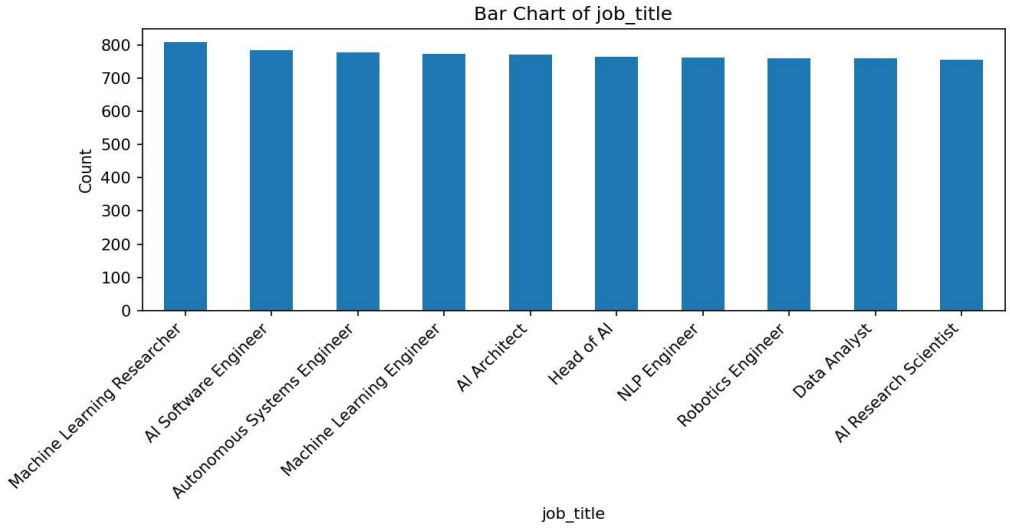
Dataset Shape After Cleaning:

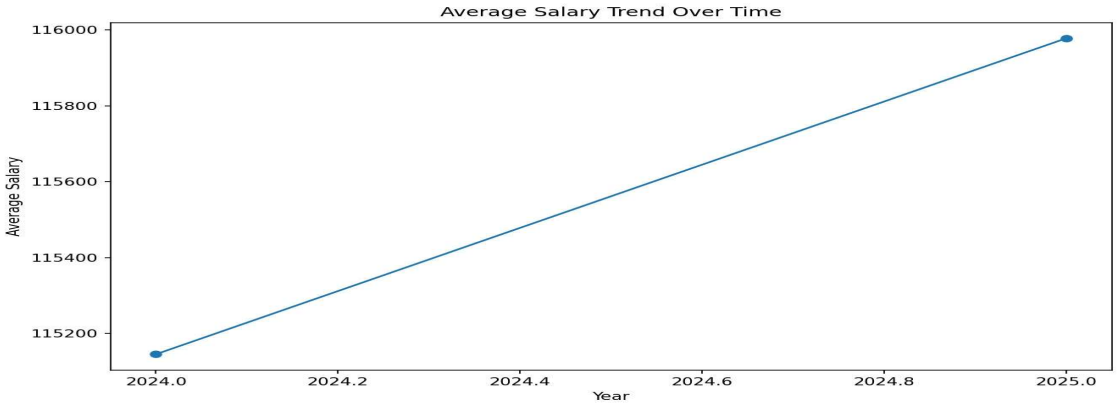
(15000, 19)

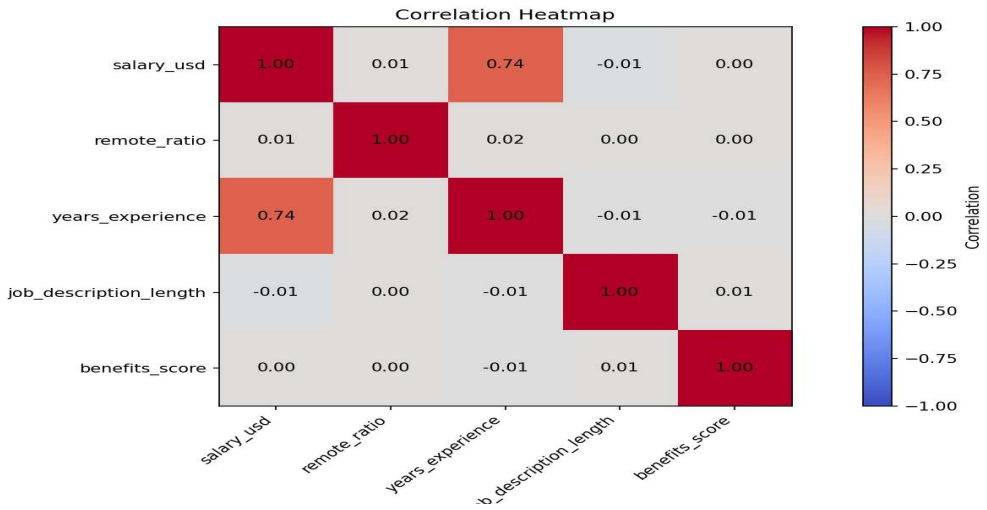
EDA COMPLETED SUCCESSFULLY




Pie Chart of job_id
AI00001
AI00006
AI00002
AI00005
AI00003
AI00004
16.7%
16.7%
16.7%
16.7%
16.7%
16.7%


Bar Chart of job_title
Count
800
700
600
500
400
300
200
100
0
Machine Learning Researcher
AI Software Engineer
Autonomous Systems Engineer
Machine Learning Engineer
AI Architect
Head of AI
NLP Engineer
Robotics Engineer
Data Analyst
AI Research Scientist
job_title


Average Salary Trend Over Time
Average Salary
116000
115800
115600
115400
115200
2024.0
2024.2
2024.4
2024.6
2024.8
2025.0
Year



Auto EDA Report

Categorical Variable:

Exploratory Data Analysis Report

Overview Variables Interactions Correlations Missing values Sample

Overview

Overview | Alerts 8 | Reproduction

Dataset statistics

Number of variables	19
Number of observations	15000
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	12.3 MiB
Average record size in memory	861.4 B

Variable types

Text	2
Categorical	11
Numeric	4
DateTime	2

Variables

job_title

job_title

Categorical

Distinct	20
Distinct (%)	0.1%
Missing	0
Missing (%)	0.0%
Memory size	980.1 KiB



More details

More details

Overview | Categories | Words | Characters

Length

Max length	27
Median length	22
Mean length	17.902267
Min length	10

Characters and Unicode

Total characters	268534
Distinct characters	35
Distinct categories	1
Distinct scripts	1
Distinct blocks	1

The Unicode Standard assigns character properties to each code point, which can be used to analyse textual variables.

Unique

Unique	0
Unique (%)	0.0%

Sample

1st row	AI Research Scientist
2nd row	AI Software Engineer
3rd row	AI Specialist
4th row	NLP Engineer
5th row	AI Consultant

Numerical Variable:

Variables

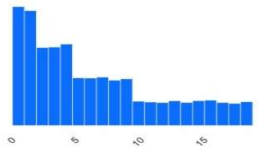
years_experience

years_experience

Real number (ℝ)

High correlation | Zeros

Distinct	20	**Minimum**	0
Distinct (%)	0.1%	**Maximum**	19
Missing	0	**Zeros**	1890
Missing (%)	0.0%	**Zeros (%)**	12.6%
Infinite	0	**Negative**	0
Infinite (%)	0.0%	**Negative (%)**	0.0%
Mean	6.2532	**Memory size**	117.3 KiB



More details

Statistics | Histogram | Common values | Extreme values

Quantile statistics

Minimum	0
5-th percentile	0
Q1	2
median	5
Q3	10
95-th percentile	17
Maximum	19
Range	19
Interquartile range (IQR)	8

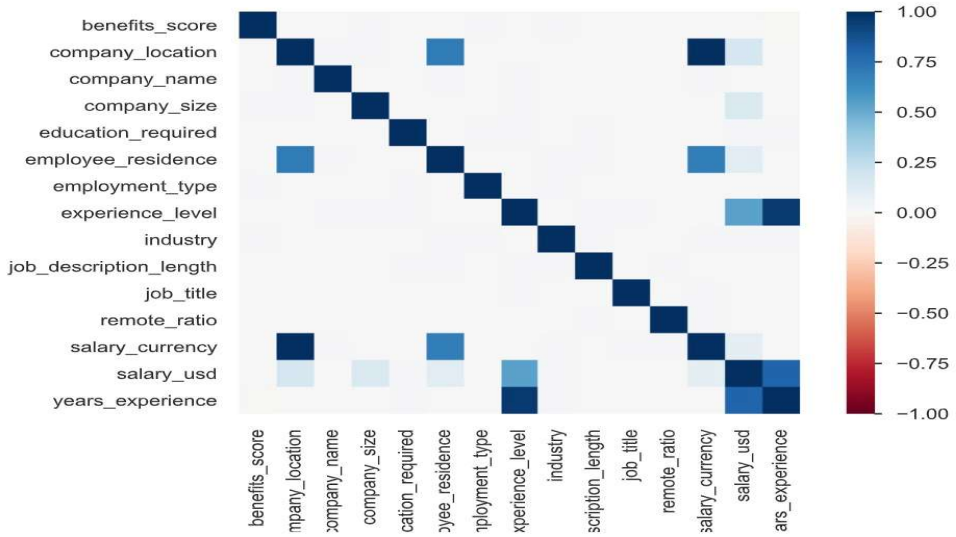
Descriptive statistics

Standard deviation	5.5457678
Coefficient of variation (CV)	0.88686877
Kurtosis	-0.52495599
Mean	6.2532
Median Absolute Deviation (MAD)	4
Skewness	0.77881321
Sum	93798
Variance	30.75554
Monotonicity	Not monotonic

Correlations

Auto

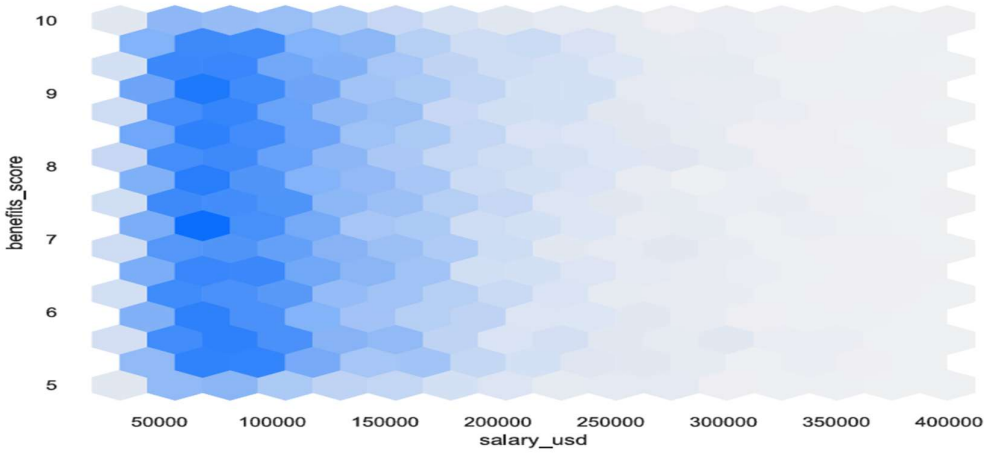
Heatmap | Table



Interactions

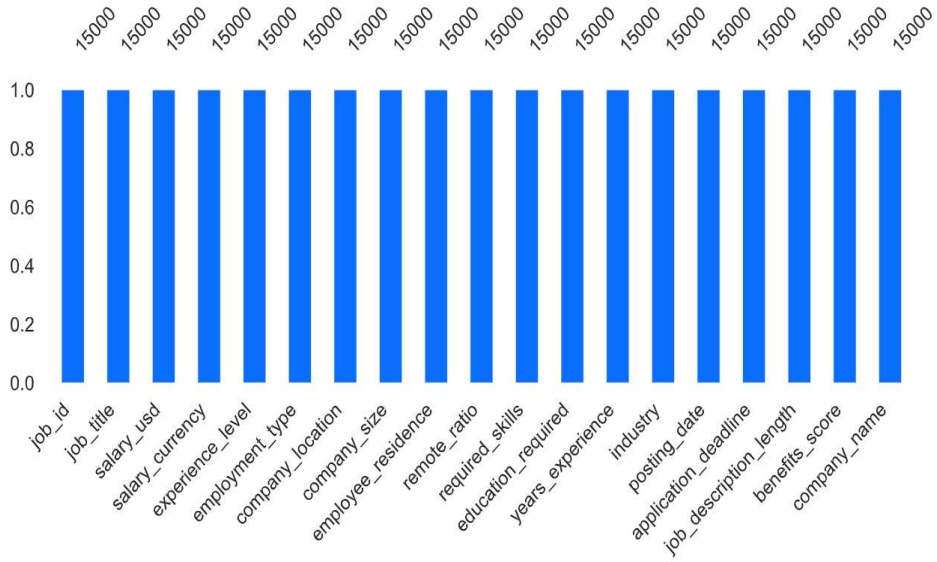
salary_usd | years_experience | job_description_length | benefits_score

benefits_score | salary_usd | years_experience | job_description_length



Missing values

Count | Matrix



A simple visualization of nullity by column.

Sample

First rows | Last rows

	job_id	job_title	salary_usd	salary_currency	experience_level	employment_type	company_location	company_size	e
0	AI00001	AI Research Scientist	90376	USD	SE	CT	China	M	C
1	AI00002	AI Software Engineer	61895	USD	EN	CT	Canada	M	Ir
2	AI00003	AI Specialist	152626	USD	MI	FL	Switzerland	L	S
3	AI00004	NLP Engineer	80215	USD	SE	FL	India	M	Ir
4	AI00005	AI Consultant	54624	EUR	EN	PT	France	S	Si
5	AI00006	AI Architect	123574	EUR	SE	CT	Germany	M	G
6	AI00007	Principal Data Scientist	79670	GBP	MI	FL	United Kingdom	S	U
7	AI00008	NLP Engineer	70640	EUR	EN	FL	France	L	Fr
8	AI00009	Data Analyst	160710	USD	SE	CT	Singapore	L	Si
9	AI00010	AI Software Engineer	102557	USD	SE	PT	Austria	M	A

Overview | Alerts 8 | Reproduction

Alerts

company_location is highly overall correlated with employee_residence and 1 other fields	High correlation
employee_residence is highly overall correlated with company_location and 1 other fields	High correlation
experience_level is highly overall correlated with salary_usd and 1 other fields	High correlation
salary_currency is highly overall correlated with company_location and 1 other fields	High correlation
salary_usd is highly overall correlated with experience_level and 1 other fields	High correlation
years_experience is highly overall correlated with experience_level and 1 other fields	High correlation
job_id has unique values	Unique
years_experience has 1890 (12.6%) zeros	Zeros

CONCLUSION

In this project, we performed Exploratory Data Analysis (EDA) on the AI Job Market and Salary Trends 2025 dataset using Python libraries such as Pandas, NumPy, Matplotlib, and YData Profiling. We analyzed job roles, salaries, skills, and employment trends using statistical summaries and graphs. The dataset contains 15,000 records and 19 columns, with no missing values or duplicate rows reported. This project helped us understand data cleaning, visualization, and analysis using Python.